

THE RESEARCH PROJECT DATA SCHEMA VOCABULARY CHALLENGE

A unified encyclopedia of scientific terminology is slowly emerging, but the FAIR-data ambition of shareable, machine-readable research still requires a provisional approach.

Scattered across sources

Ontologies, SKOS vocabularies (AGROVOC), portals (BioPortal, OLS4), plain web pages or PDFs — no two expose terms the same way.

Irregularly shaped

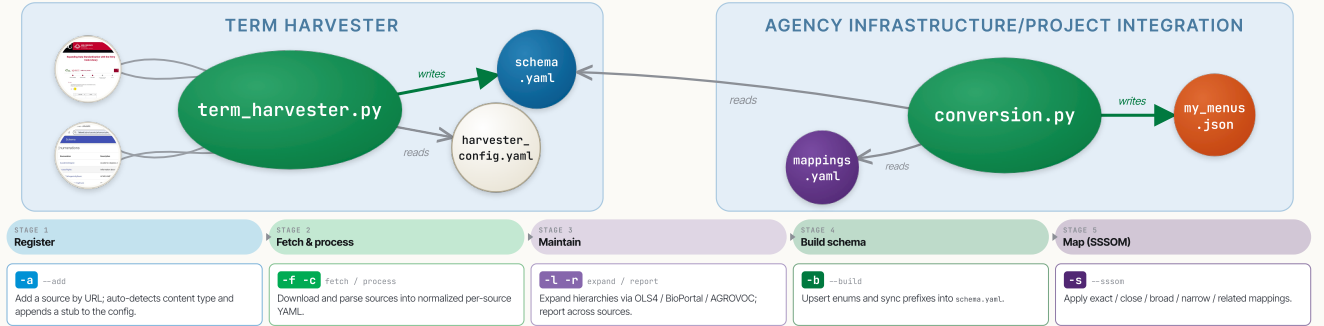
Hierarchies, enum picklists, inclusions/exclusions, and obsolete/deprecated status are each encoded differently.

Often evolving

Vocabularies are versioned and revised; keeping a specification current is recurring, consensus-driven work.



THE FLOW SOLUTION



SOURCE CONNECTORS (HIGHLIGHTS)

OntologyAPI

No file is downloaded — the api block routes CURIES to OLS4, BioPortal, or AGROVOC; -l expands into the relevant subtree.

PARSED FRAGMENT - AGROVOC_C_2808.YAML

```
enums:
  AGROVOC_C_2808:
    title: formand
    permissible_values:
      agrovoc_c_18948:
        title: available land
        is_a: agrovoc_c_2888
```

STABLE RESOURCE - SUBTREE

```
python term_harvester.py -a SMO-080000000
python term_harvester.py -l SMO-080000000
```

WHOLE RESOURCE - N/A

API-driven by design; add entity concepts, let -l walk only their subtrees — a full ontology dump defeats the mission.

OLS

Home Ontology Search

Welcome to the EMBL-EB

Search Tag Text

Search OLS...

Get match Include

NSDB

AAFC's National Soil Database, scraped from HTML attribute pages — each table contributes one enum group, bilingual ENFR.

PARSED FRAGMENT - NSDB001.WATERBIL.YAML

```
enums:
  NSDB_WATERBIL:
    title: Water Table Characteristics
    permissible_values:
      W:
        title: Always
      N:
        title: Never
```

STABLE ATTRIBUTE PAGE

```
python term_harvester.py -a ".../nsdb/waterbil/001/index.html"
```

WHOLE RESOURCE

Top-level NSDB index registers the combined Soil Name + Layer + one source.

```
python term_harvester.py -a ".../nsdb/001/02/index.html"
```

NASIS

USDA NRCS domain tables from one massive PDF — NASIS 7.4x yields ~453 enums across 906 pages; concise: true drops obsolete-marked PVL.

PARSED FRAGMENT - NRCS001FIELDBOOK.YAML

```
enums:
  .SalinityClass:
    title: Salinity Class
    permissible_values:
      N:
        title: non saline
      S:
        title: strongly saline
```

ONE DOMAIN BY NAME

```
python term_harvester.py -c NASIS -b
```

WHOLE RESOURCE

Single -a registers the PDF, saves it, and runs the parser for every NASIS domain.

```
python term_harvester.py -a "NASIS_7.4_1_Salinity.pdf"
```

FreeText

Configure Term Harvester to capture a section of text from a pdf or web page or text document, and then send it to the Anthropic API for text analysis; extracted values are provided for review before acceptance.

CLAUDE-EXTRACTED - LODGINGSCALE.YAML

```
enums:
  LodgingScale:
    title: Lodging Severity Scale
    permissible_values:
      1:
        title: none
      2:
        title: severe
```

ADD FROM A PDF OR TEXT

```
python term_harvester.py -a "field-guide.pdf" --free-text "lodging 3-9"
```

WHOLE RESOURCE - REFRESH

-f downloads the doc, then -e re-extracts and diffs vs. the saved YAML.

LODGING

A visual estimation of lodging before harvest was done using a traditional scale of 1-9, similar to that described by Berry et al [25]. On that scale, 9 means "no lodging", 7- plants are slightly lodged or stand upright after lodging, 5- medium degree of lodging, 3- plants lodged significantly making machine harvesting difficult, 1- severe lodging making harvesting impossible. Other values denote intermediate states of lodging.

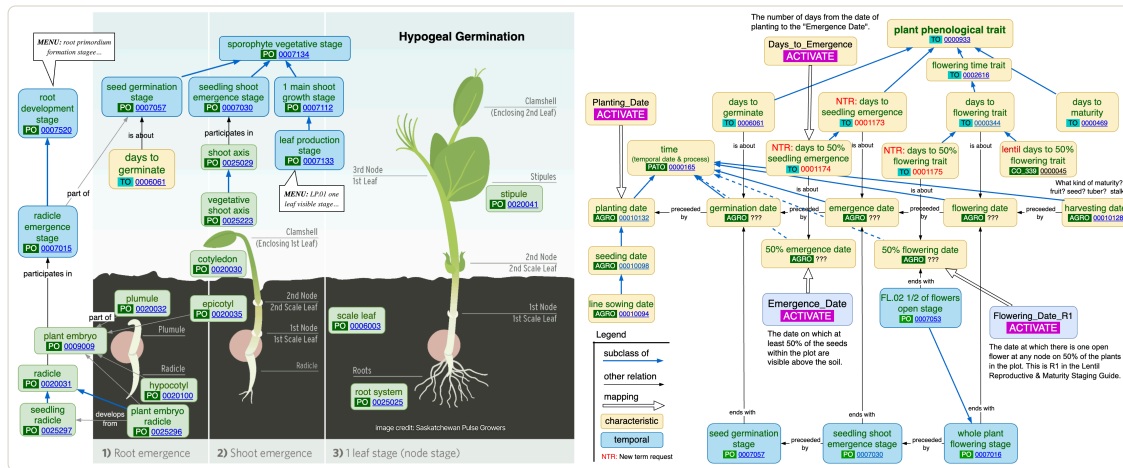
ALL SOURCE CONNECTORS: LinkML (raw.githubusercontent.com/LinkML/valuexists/.../merged_hierarchy.yaml), OntologyAPI (OLS4 / BioPortal / AGROVOC REST & SPARQL endpoints), OWL (gutenberg.org/obo/ene.owl), LOINC (loinc.org/en/valuexists.html), LOINCCodeSystem/ValueSet (terminology.hl7.org/7.1.1/en/codeSystem-data-absent-reason.json), NSDB family (sis.agr.ca/canisis/nadb/001/02/index.html), STATSON (we3.statcan.ca/sum/p39/p31Function/getP670-1368854), NAPCSCanada (statcan.ca/en/nea/5274), AgFoodCA (github.com/agricofoodcanada/picklists, fr_schemas), NASIS / NRCS Field Book (nrcs.usda.gov/.../NASIS_7.4_1_Salinity.pdf), FreeText (user-supplied PDF, text, or URL), ISO_COUNTRY (iso.org/obp/ul/iso-country:3364.ca via Wikidata SPARQL)

Ontology: The semantic nexus of controlled vocabulary

Domain-specific ontologies have the potential to act as translational fulcrums for the various terminology sources.

- The OBO Foundry¹ family of life and natural science ontologies is a coordinated encyclopedia of taxonomy, chemistry, genomics, anatomy, disease, plant biology, agriculture, food system, and broader environmental and organismal health terms.
- An upper level ontology framework imposes a logical structure of categories within which the foundry ontologies mostly conform: **material entities**, their **characteristics** and **roles, processes, and information** (data types). A shared set of relations for describing content and parthood (in the Relations Ontology) enables consistent knowledge graph construction.
- The precision and consistency of this framework is evolving, thanks to the support of various institutions, grants, and volunteers.
- We also use the LinkML² data structure framework for holding all controlled vocabulary, and mapping-commons.github.io/sssom for establishing term mappings.

¹Rebecca Jackson et al., *OBO Foundry in 2021: operationalizing open data principles to evaluate ontologies*, Database, Volume 2021, 2021, baab069, <https://doi.org/10.1093/database/baab069>



CASE STUDY

Mission: to have a Term Harvester configuration retrieve the Plant Trait Ontology **root development stage** [PO:0007520] menu and the **leaf production stage** [PO:0007172] menu.

`python3 ../term-harvester/term_harvester.py -a PO:0007520,PO:0007172`

HARVESTER_CONFIG.YAML

```
sources:
  PO_0800733:
    title: leaf production stage
    name: PO_0800733
    version: [...] 2026-01-09/po.owl
    content_type: OntologyAPI
    file_format: json
    resolvable_from:
      api:
        ois:
          type: root
          source_nodes:
            - PO_0800733
          include_self: true
          download_date: '2026-07-14'
          description: This stage is recommended over
            rosette growth or stem elongation as it is
            most easily observed and least ambiguous.
          prefixes:
            PO: http://purl.obolibrary.org/obo/PO/
            PO_08007520:
              title: root development stage
              name: PO_08007520
              ...
```

ONGOING WORK

To support CS-DCC projects (like the Genome Canada funded ACTWATING Genomics to Accelerate Climate-Smart Crop Delivery project) we are enhancing the Plant Trait Ontology (TO) and AGRO ontologies to support variables like "50% flowering date" and "days to 50% seedling emergence".

SCHEMA.YAML

```
id: https://example.org/plant_research
name: PlantResearchVocabulary
title: Plant Research Vocabulary
description: A selection of plantlists useful
  for plant breeding research
version: 1.0
license: CC0
antiquities: https://github.com/anthropics/
  PO: http://purl.obolibrary.org/obo/PO/
  PO_0800733:
    title: leaf production stage
    name: PO_0800733
    version: [...] 2026-01-09/po.owl
    content_type: OntologyAPI
    file_format: json
    resolvable_from:
      api:
        ois:
          type: root
          source_nodes:
            - PO_0800733
          include_self: true
          download_date: '2026-07-14'
          description: This stage is recommended over
            rosette growth or stem elongation as it is
            most easily observed and least ambiguous.
          prefixes:
            PO: http://purl.obolibrary.org/obo/PO/
            PO_08007520:
              title: root development stage
              name: PO_08007520
              ...
```